SUMMARY TABLE: MATHEMATICS GRADE 10

Sub-Strand 2.3: Area of Part of a Circle — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 2.3: Area of Part of a Circle
Driving Question	DRIVING QUESTION: How can we use the radius and the central angle of a circular pizza slice to calculate its exact area — and apply the same method to solve real problems in farming, design, and everyday life in Kenya?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 **Anchoring Phenomenon & Predict — Unequal Slices, Unequal Areas**	Students examined a circular pizza (or chapati) cut into unequal slices and observed that slices with larger central angles visibly occupied more area than smaller ones. They noted that the full circle has a central angle of 360° and that each slice appeared to be a proportional fraction of the whole.	Students learned that a sector is a 'pizza-slice' region of a circle bounded by two radii and an arc, and that its area is a fraction of the total circle area — specifically the fraction (θ/360°). They were introduced to the conjecture that Area of sector = (θ/360°) × πr^2, which they will prove in later lessons.	The teacher should use a physical chapati or draw a circle on the board divided into unequal sectors. Ask students to rank slices by area before any formula is given — this surfaces prior knowledge and misconceptions. Pedagogical note: cold-call at least 4 students to justify their ranking; allow 30 seconds of wait time before accepting answers. Record all predictions on the Driving Question Board (DQB) so they can be revisited and revised throughout the unit. The goal of this lesson is not correctness but productive curiosity — students should leave wondering HOW we calculate the exact area.
Lesson 2 **OBSERVE & REVIEW — Parts of a Circle, Radius, Central Angle, and the Hunt for a Pattern**	Students labelled a diagram of a circle identifying: centre, radius, diameter, chord, arc (minor and major), sector (minor and major), segment (minor and major), and central angle. They then measured and recorded the central angles and estimated areas of several pre-drawn sectors, hunting for a numerical pattern between angle size and area.	Students consolidated vocabulary for all parts of a circle and confirmed through data collection that as the central angle doubles, the sector area also doubles — pointing directly to a proportional (linear) relationship between θ and area. They also reinforced that a full circle (θ = 360°) gives the familiar area formula πr^2.	The teacher should distribute circle-labelling worksheets and have students work in pairs before whole-class discussion. Pedagogical note: deliberately include a 'segment' and a 'sector' side-by-side on the diagram — students frequently confuse these two terms, and early clarification is critical for Lessons 6 and 7. When reviewing the pattern hunt, ask: 'If a sector with θ = 90° has area X, what area do you predict for θ =

			180°?' Cold-call 3 students and record responses on the DQB. Emphasise that this lesson is gathering EVIDENCE to later justify the formula — connect explicitly back to the chapati anchoring phenomenon.
Lesson 3 **EXPLORE — Constructing Sectors and Measuring Area by Counting Unit Squares**	Students used a compass and protractor to construct sectors of given radii and central angles on squared paper, then estimated each sector's area by counting unit squares inside the boundary. They compared their counted estimates with the prediction from the formula $(\theta/360°) \times \pi r^2$ to assess how closely the formula matched reality.	Students learned that the formula $(\theta/360°) \times \pi r^2$ is not merely a rule to memorise but an accurate model that closely predicts the physical area of a drawn sector. Minor discrepancies arise from the curved boundary cutting through unit squares, reinforcing why an exact algebraic formula is more reliable than counting. This lesson bridges the visual/intuitive understanding from Lessons 1–2 to the formal derivation in Lesson 4.	The teacher should prepare squared paper with circles of radius 6 cm pre-drawn (to save time). Assign different central angles to different groups (e.g., 60°, 90°, 120°, 150°) so that results can be pooled for a class data table. Pedagogical note: after counting, have each group calculate $(\theta/360°) \times \pi r^2$ and compute the percentage error — this embeds numeracy and critical thinking. Ask: 'Why is our counted value slightly off? Is the formula wrong, or is counting on a grid imprecise?' Allow 45 seconds of think time before discussion. Connect back to the DQB: update any predictions students made in Lesson 1.
Lesson 4 **EXPLAIN — Deriving and Applying the Area of a Sector (Minor Sector)**	Students followed a step-by-step derivation showing that a sector is a fraction $(\theta/360°)$ of the full circle, so its area is that same fraction of πr^2. They applied the formula $A = (\theta/360°) \times \pi r^2$ to calculate sector areas for a chapati divided among family members, a maize farm plot shaped as a sector near Kericho, and a decorative tile design in Mombasa.	Students derived and can fluently apply the sector area formula $A = (\theta/360°) \times \pi r^2$. They understand that the formula works for both minor sectors ($\theta < 180°$) and major sectors ($\theta > 180°$) because it is purely proportional. They also learned to reverse the formula to find r or θ when area is given — an important algebraic manipulation skill.	The teacher should write the derivation on the board in three logical steps: (1) full circle area = πr^2, (2) sector is $\theta/360°$ of the full circle, (3) therefore sector area = $(\theta/360°) \times \pi r^2$. Pedagogical note: after deriving, immediately pose a reverse problem — 'A sector has area 38.5 cm² and radius 7 cm; find θ.' Cold-call 2 students to set up the equation before the class solves it. Emphasise correct use of π (use $\pi = 22/7$ or 3.14 as directed by KICD) and consistent units. This is the CORE lesson of the sub-strand; ensure all students complete at least three worked examples in their exercise books before the lesson ends.
Lesson 5 **PRACTICE & CONSOLIDATION — Sector Area in Kenyan Real-Life Contexts**	Students worked through a structured set of problems set entirely in Kenyan contexts: calculating the area of a sector-shaped shamba (farm) near Lake Victoria given its radius and central angle, finding the area of a slice of watermelon, and determining how much paint is needed to cover a sector-shaped wall mural in a Nairobi school.	Students consolidated fluency with $A = (\theta/360°) \times \pi r^2$ across varied contexts, including finding area when r and θ are given, finding r when area and θ are given, and finding θ when area and r are given. They also practised converting between different units (e.g., metres to centimetres) before substituting into the formula — a common source of error identified in earlier	The teacher should use a 'gradual release' structure: one fully worked example together, one example completed in pairs, then independent practice. Pedagogical note: circulate during independent practice and look specifically for two common errors — (1) students using the diameter instead of the radius, and (2) students forgetting to multiply by π. Address both errors whole-

		lessons.	class with a brief 5-minute error-analysis discussion at the end. Connect to the DQB: ask students whether any of their original Lesson 1 predictions have now been confirmed or refuted.
Lesson 6 **Area of a Minor Segment — The Watermelon Rind Problem**	Students examined a cross-section of a watermelon slice and identified the minor segment as the region between a chord and its minor arc — distinct from the sector. They observed that removing the triangular region (the triangle formed by the two radii and the chord) from the sector leaves exactly the segment, leading to the subtraction formula.	Students derived and applied the formula: Area of minor segment = $(\theta/360°) \times \pi r^2 - \frac{1}{2}r^2\sin\theta$. They understand that the sector area gives the full 'pie-slice' region and that subtracting the area of the isosceles triangle ($\frac{1}{2}r^2\sin\theta$) isolates the curved segment. They can correctly identify which part of a diagram is the segment versus the sector.	The teacher should draw a large sector on the board, shade the triangle in one colour and the segment in another, and ask: 'How do we find ONLY the shaded curved part?' This visual makes the subtraction relationship self-evident. Pedagogical note: derive $\frac{1}{2}r^2\sin\theta$ from the standard triangle area formula (½ × base × height), substituting r for both sides of the isosceles triangle — do not present it as a 'given' formula. Cold-call 3 students to explain each step of the derivation in their own words. The watermelon rind context grounds the abstract concept; refer back to it throughout. Note: students must know how to use the sine function on a calculator — briefly verify this skill at the start of class.
Lesson 7 **Area of a Major Segment — Circular Plot Allocation in the Rift Valley**	Students examined a circular land plot near Nakuru in the Rift Valley divided by a chord into a minor and a major segment. They observed that the two segments together make the full circle, so the major segment area must equal πr^2 minus the minor segment area. This complementary relationship was verified using actual numbers.	Students learned and applied: Area of major segment = πr^2 − Area of minor segment, where Area of minor segment = $(\theta/360°) \times \pi r^2 - \frac{1}{2}r^2\sin\theta$, and θ is the central angle subtended by the MINOR segment. They can correctly identify which angle to use and understand that the same chord defines both segments simultaneously.	The teacher should begin by asking: 'If the minor segment has area X and the full circle has area πr^2, what is the major segment area?' — this elicits the complementary relationship without giving away the formula. Pedagogical note: the most common error in this lesson is students using the reflex angle (the major angle) in the minor segment formula instead of the minor central angle θ. Prepare a deliberate 'wrong solution' on the board and ask students to find the error — this is more effective than simply warning them. Connect to the Rift Valley farming context throughout: frame the problem as a farmer needing to know how much of a circular irrigation plot falls on each side of a dividing fence (chord).
Lesson 8 **Application & Model Building — Solving Real-Life Problems Using Arc Length, Sector Area, and**	Students tackled multi-step real-life problems that required selecting and combining the correct formula from arc length $(\theta/360° \times 2\pi r)$, sector area $(\theta/360° \times \pi r^2)$, and segment area formulas. Problems	Students learned that all three formulas (arc length, sector area, segment area) derive from the same foundational idea — a proportional fraction of the full circle — and that choosing the right formula depends on	The teacher should present each problem WITHOUT the formula pre-labelled and ask students to first identify: 'Are we measuring a LENGTH or an AREA? Is the region a sector or a segment?' This metacognitive

Segment Area	included fencing a sector-shaped school garden in Kisumu, tiling a decorative segment-shaped panel for a Mombasa hotel, and calculating the irrigated area of a pivot-irrigation farm.	what region or boundary is being measured. They practised reading problem contexts carefully to determine whether a sector, segment, arc, or combination is required.	step prevents mechanical formula-grabbing. Pedagogical note: use group work (groups of 3–4) for this lesson, assigning one problem per group, then having groups present solutions. Require each group to draw a labelled diagram before writing any formula — diagrams dramatically reduce errors. Update the DQB at the end of this lesson: students should now be able to answer the driving question fully for sectors and partially for segments.
Lesson 9 **Digital Tools & Consolidation — Confirming Sector and Segment Areas, Correcting Errors**	Students used GeoGebra (or a similar digital tool available offline on the ARES platform) to construct sectors and segments, dynamically vary the central angle and radius, and observe how the computed area changed. They also analysed pre-prepared 'student solutions' containing deliberate errors — including the degree/radian confusion and diameter-for-radius substitution — and corrected them.	Students learned that the sector area formula $A = (\theta/360°) \times \pi r^2$ requires θ to be in degrees; using radians without adjustment produces incorrect answers. They also reinforced that r is the radius (half the diameter) and practised checking reasonableness of answers (e.g., a sector area cannot exceed πr^2). Error-analysis tasks deepened conceptual understanding beyond mechanical application.	The teacher should project the GeoGebra applet (pre-loaded offline) and demonstrate how dragging the angle slider changes the sector area in real time, asking: 'What happens to the area when θ = 360°? Does it match πr^2?' This dynamic visualisation consolidates the proportional reasoning developed across the unit. Pedagogical note: the error-correction task is best done individually first (5 minutes silent), then discussed in pairs, then whole-class — this structure ensures every student attempts diagnosis before hearing peers' reasoning. Emphasise to students that professional engineers and architects in Kenya use exactly these tools and checks on real projects.
Lesson 10 **Assessment & Reflection — Cumulative Model Presentation and Driving Question Resolution**	Students presented their cumulative models (diagrams, annotated formulas, and real-life examples built up across Lessons 1–9) and answered the driving question in full: explaining how the radius and central angle of a pizza/chapati slice determine its exact area, and how the same logic applies to farming plots, design, and everyday life in Kenya.	Students demonstrated mastery of the full progression: from the anchoring phenomenon (unequal chapati slices) to the sector area formula, to minor and major segment areas, to multi-step real-life applications. They can articulate WHY the formula works (proportional fraction of the full circle) and not just HOW to apply it — the hallmark of competency-based understanding as required by the KICD Grade 10 curriculum.	The teacher should structure this lesson as a 'gallery walk' or brief presentation (2–3 minutes per group) where each group shares one real-life problem they solved using the formulas from this sub-strand. Pedagogical note: use a structured peer-feedback protocol — each listening group must identify (1) one thing explained clearly and (2) one question they still have. Close by returning to the DQB from Lesson 1: read aloud the original student predictions and ask the class to vote on which have been confirmed, which were misconceptions, and which are still open questions. This metacognitive closure is essential in CBE — it makes the learning journey visible to students and the teacher alike.